Practice Quiz 09B — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 09

An image counterexample; completing a geometric-sum induction step.

1. (a) Construct a function $f : \{1, 2\} \to \{a\}$ and choose subsets $W, X \subseteq \{1, 2\}$ for which

$$f(W \setminus X) \neq f(W) \setminus f(X).$$

Verify your counterexample by computing both sides.

Solution Define $f(1) = f(2) = a$, and choose $W = \{1\}$ and $X = \{2\}$. Then $W \setminus X = \{1\}$, so

$$f(W \setminus X) = \{a\}.$$

But $f(W) = f(X) = \{a\}$, and therefore

$$f(W) \setminus f(X) = \{a\} \setminus \{a\} = \varnothing.$$

The two sets are different, as required. ∎

(b) Fix a real $r \neq 1$ and an integer $n \geq 0$. Assume

$$\sum_{i=0}^{n} r^i = \frac{r^{n+1} - 1}{r - 1}.$$

Complete the induction step by proving the corresponding formula for $n + 1$. Explicitly identify the new term.

Solution The new term is r^{n+1}. By the given inductive hypothesis,

$$\begin{aligned}\sum_{i=0}^{n+1} r^i &= \left(\sum_{i=0}^{n} r^i\right) + r^{n+1} \\ &= \frac{r^{n+1} - 1}{r - 1} + r^{n+1} \\ &= \frac{r^{n+1} - 1 + r^{n+2} - r^{n+1}}{r - 1} = \frac{r^{n+2} - 1}{r - 1}.\end{aligned}$$

This is exactly the required formula with $n + 1$ in place of n. ∎

Solutions revised using GPT-6 Astra in the Codex harness.